

EDUCATION

Master of Science in Applied Mathematics , Florida Atlantic University, Boca Raton, FL	2026–2027
- GPA: 4.00/4.00 - Thesis: The Topology of Optimizers Beyond the Edge of Stability - Advisor: Dr. Jason Mireles-James	
Bachelor of Science in Mathematics with Honors , Florida Atlantic University, Boca Raton, FL	2023–2026
GPA: 4.00/4.00	
High School Diploma , Florida Atlantic University High School, Boca Raton, FL	2023–2027
- GPA: 4.00/4.00 - Concurrently dual-enrolled in combined BS-MS program	

AWARDS & HONORS

Barry M. Goldwater Scholar (\$7500), Florida Atlantic University	March 2026
Phi Kappa Phi Honor Society Scholarship (\$1000), Florida Atlantic University	March 2026
Joint Mathematics Meetings Travel Grants (\$250+\$800), Emory University	January 2026
OURI Research Grant (\$600), Florida Atlantic University	January 2026
Florida Launch Scholar (\$610), Florida Atlantic University	October 2025
Mu Alpha Theta Summer Grant (\$2000), Florida Atlantic University	June 2024

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA	
<i>SIG Research Fellow</i>	Jul.–Aug. 2026
- Selected for competitive six-week research program in geometry processing; completed three two-week projects with researchers and industry professionals - Topics: fluid dynamics, computer graphics, computer vision, discrete differential geometry <i>Fluid Dynamics on Curved Surfaces (mentor: Dr. Mohammed Sina Nabizadeh):</i> - Developed Discrete Exterior Calculus-based solvers for incompressible Euler flow on planar and curved triangular meshes, incorporating Hodge projection and Clebsch representations - Evaluated stability and mass conservation using perturbed Taylor-Green vortices and Laplace-Beltrami flows - Simulated Eulerian flows on Polyscope and Blender - Ongoing research; [blog], [supplementary] in progress. <i>Self-Rectifying Textures (mentor: Dr. Max Dunitz):</i> - Constructed/deformed textures using homographies and cylindrical warps with OpenCV. - Developed Minimum Spanning Tree and Finite Element Method-based rectification pipeline to recover inverse deformation mappings. - Work documented in [blog]. <i>Locality in Geometric Algorithms (mentors: Drs. Sebastien Loriot and Mael Rouxel-Labbé):</i> - Re-implemented Morton, Tipsify, and Hilbert sorting algorithms in C++. - Compared runtime of accelerated Edge Collapse algorithm under Morton-sorting, Tipsify-sorting, and Hilbert-sorting against randomly ordered vertices. - Work documented in [blog].	

Florida Atlantic University, Boca Raton, FL

Undergraduate Researcher (Advisor: Dr. Yu Xiang)

2026–Present

- Topics: conformal inference, statistical learning, novelty detection, information theory
- Developed algorithm for conformal novelty detection with finite-sample false discovery rate (FDR) guarantees under a decentralized composite null hypothesis with limited communication.
- Rigorously bounded FDR of my algorithm and empirically demonstrated stronger statistical power and FDR control on synthetic Gaussian numerical experiment in Python.

Emory University, Atlanta, GA

Computational Mathematics REU (Advisor: Dr. Matthias Chung)

Jun.–Aug. 2025

- Topics: scientific machine learning, inverse problems, multivariate analysis
- Derived optimal rank-constrained linear encoder-decoder architectures for rank-deficient data and operators.
- Demonstrated competitive performance for inverse recovery tasks on finance and shallow water PDE datasets using PyTorch.
- Contributed theorem development, numerical experiments, and implementation for a collaborative project published in *Foundations of Data Science*.

Florida Atlantic University, Boca Raton, FL

Master's Thesis (Advisor: Dr. Jason Mireles-James)

2025–Present

- Topics: deep learning optimization, dynamical systems, computer-assisted proofs
- Derived lower bound for learning rates in which RMSProp and Adam optimization algorithms are Lyapunov unstable and upper bound for attractor size.
- Conducted bifurcation analysis of RMSProp in high learning-rate regimes, characterizing strange attractors and computing the parametrizations and invariant bundles of period-2 invariant tori using MATLAB.
- Validated period-2 orbits, Neimark-Sacker bifurcations, normally hyperbolic quasiperiodic behavior using computer-assisted proofs and interval arithmetic with INTLAB and Julia.

TEACHING & OUTREACH

Florida Atlantic University High School, Boca Raton, FL

Mathematics Tutor

2023–present

- Tutored 20 students across 400 hours of volunteer tutoring in Algebra II, Precalculus, Calculus, Biology, Physics.
- Mentored 6 students in advanced undergraduate-level mathematics and computer science, providing one-on-one guidance on course selection, internship application essay-writing, connections with faculty in mathematics department.
- Publicized biology lecture notes for 9th grade honors biology course [[link](#)].
- Delivered one-on-one and small-group instruction both in person and remotely.

Kyle Loh's Math, YouTube

Founder and Content Creator

2022–Present

- Produced undergraduate-level mathematics tutorials with emphasis on differential equations, dynamical systems, and GRE Mathematics Subject topics [[link](#)].
 - Developed 137 tutorials reaching approximately 94,000 cumulative views and 730 subscribers.
 - Created supplementary instructional tutorials on Google Colab following my ongoing series “Dynamics of Learning Algorithms” designed to improve accessibility of research-level mathematics [[link](#)].
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PUBLICATIONS & PRESENTATIONS

Refereed Articles & Preprints

- A. DeLise, **K. Loh**, K. Patel, M. Teague, A. Arnold, M. Chung. "Optimal Linear Baseline Models for Scientific Machine Learning." American Institute of Mathematical Sciences, Foundations of Data Science, 2026. [[doi](#)] [[arXiv](#)]
- **K. Loh**, Y. Xiang. "Decentralized Conformal Novelty Detection via Quantized Model Exchange," 2026. [[arXiv](#)]

Poster Presentations

- Apr 2026. FAU Undergraduate Research Symposium, Florida Atlantic University, Boca Raton, FL. The Convergence Behavior of RMSProp.
- Jan 2026. Joint Mathematics Meetings AMS-PME Student Poster Session, Washington D.C., Optimal Rank-Constrained Mappings for Linear ED Architectures [[pdf](#)].
- Aug 2025. CMDS REU Poster Presentation, Emory University, Atlanta, GA. Optimal Rank-Constrained Mappings for Linear ED Architectures [[pdf](#)].

PROFESSIONAL MEMBERSHIPS

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| • Phi Kappa Phi Honors Society | Mar. 2026 |
| • IEEE | Mar 2026 |
| • American Mathematical Society | Jan. 2026 |

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, Julia, C++, R
- **Machine Learning Frameworks:** PyTorch, TensorFlow, OpenCV
- **Simulation Software:** Blender, Simulink, FEBio
- **Other:** Git, HTML, Astro